

Parameter	Value
Base model	Qwen2.5-VL-3B-Instruct (BF16)
LoRA rank r	16
LoRA α	32
LoRA dropout	0.05
Target modules	q, k, v, o, gate, up, down proj.
Learning rate	2×10^{-5}
Epochs	40
Effective batch size	4 (grad. accumulation)
Scheduler	Cosine with 10% warmup
Gradient clipping	1.0

Table 1: LoRA hyperparameters.

Dataset	Modality	Classes	Train	Test
Granulometry [10]	Macro photo	9	18	108
NEU-CLS [11]	Grayscale	6	30	360
UHCS [12]	Microscopy	5	30	117
RIAWELC [13]	X-ray	4	24	240

Table 2: Dataset summary.

Data augmentation through reasoning diversity. With only 18–30 training images, memorization is a real risk. The 3 different descriptions per image provide variation in how visual features are described, reducing overfitting to specific image patterns. This is consistent with findings from Hsieh *et al.* [9], who showed that rationale-augmented training data helps small models generalize better than direct fine-tuning alone.

Contrastive decision boundary learning. The generated descriptions explicitly state why an image is not a similar class. With Direct LoRA, the model only learns “this image = this label.” With the augmented approach, it learns “circular dark spots = porosity, NOT cracks (which would be jagged lines).” This transfers more effectively to unseen images where the distinction between similar classes matters.

4 Experimental Setup

4.1 Datasets

We evaluate our method on four industrial classification tasks spanning three image modalities. Each task represents a real-world inspection scenario where labeled data is scarce and domain expertise is required.

Concrete Aggregate Grading (Granulometry). This dataset [10] contains macro photographs of concrete aggregate samples classified along two axes: maximum particle size (8, 16, or 32mm) and grading according to DIN 1045 [9] (coarse, medium, or fine). The combination yields 9 classes. Images are 2200×3000 pixels, resized to 800px maximum dimension during processing. We use 18 training images (2 per class) and 108 test images. The ground sampling distance (GSD) is computed dynamically based on the resize factor and used in the classification prompt.

Steel Surface Defect Detection. The NEU-CLS dataset [11] contains 200×200 pixel grayscale photographs of hot-rolled steel strip surfaces with 6 defect classes: crazing, inclusion, patches, pitted surface, rolled-in scale, and scratches. We use 30 training images (5 per class) and 360 test images from the validation split.

Steel Microstructure Classification (UHCS). The NIST Ultra-High Carbon Steel dataset [12] contains optical/SEM micrographs at varying magnifications showing microstructural features of UHCS after heat treatment. We classify 5 microconstituent classes: spheroidite, network, spheroidite+widmanstatten, pearlite+spheroidite, and pearlite. Images are 645×481 pixels. We use 30 training images (6 per class) and 117 test images. Magnification metadata is included in the classification prompt where available.

Weld Defect Classification. The RIAWELC dataset [13] contains 227×227 pixel grayscale